



3.34.1 Serial wire JTAG debug port (SWJ-DP)

The Arm® SWJ-DP interface is embedded, and is a combined JTAG and serial wire debug port that enables either a serial wire debug or a JTAG probe to be connected to the target.

Debug is performed using 2 pins only instead of 5 required by the JTAG (JTAG pins could be re-use as GPIO with alternate function): the JTAG TMS and TCK pins are shared with SWDIO and SWCLK, respectively, and a specific sequence on the TMS pin is used to switch between JTAG-DP and SW-DP

Power supply management: [datasheet](#) p.20

Figure 2. Power supply overview

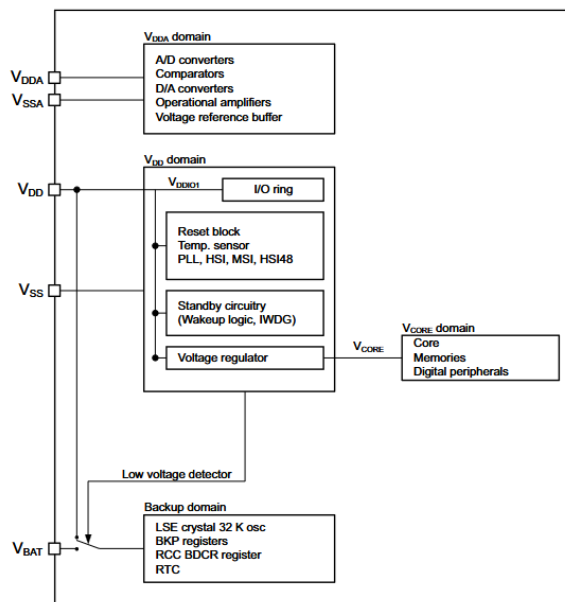


Figure 1. STM32L431xx block diagram

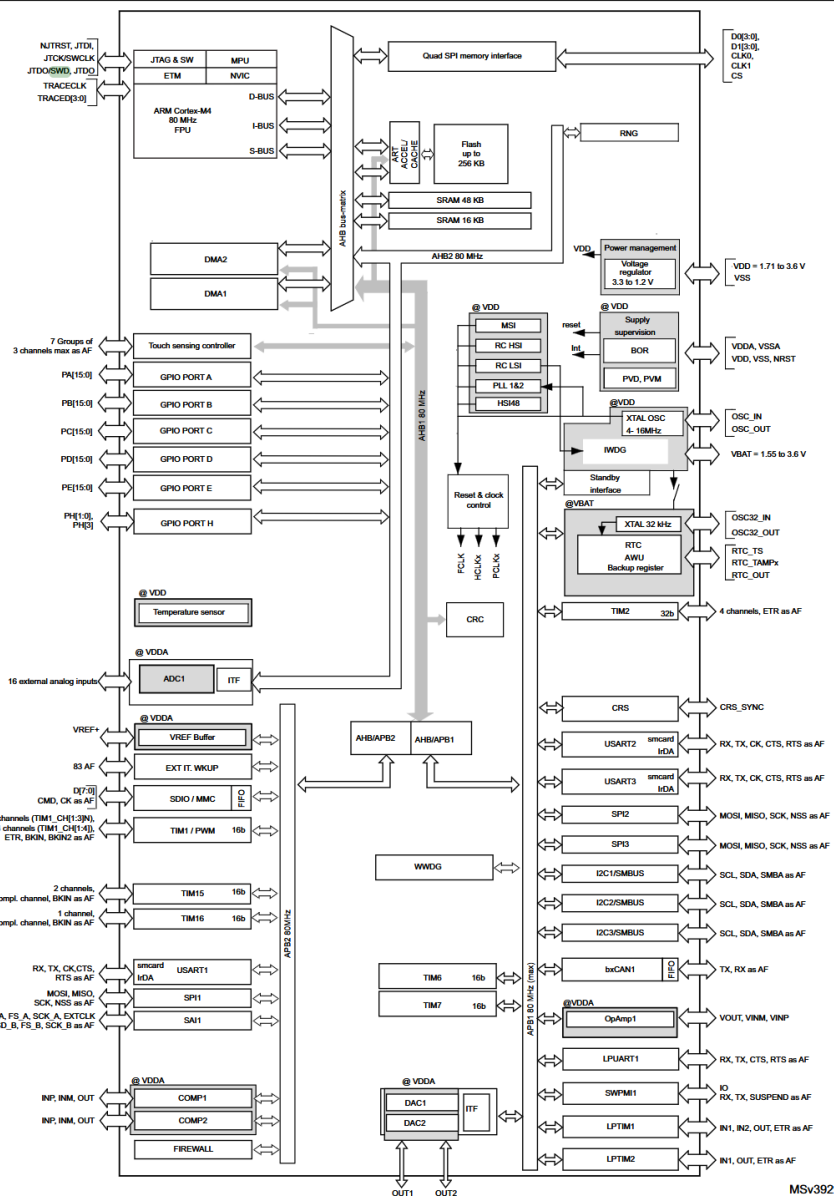
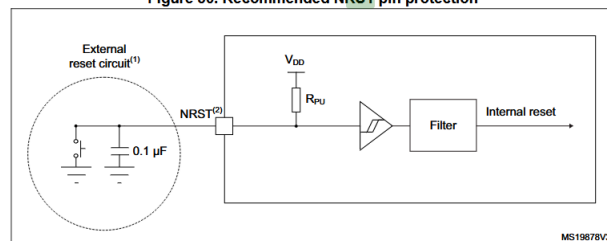
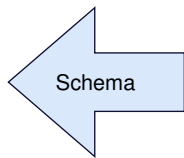
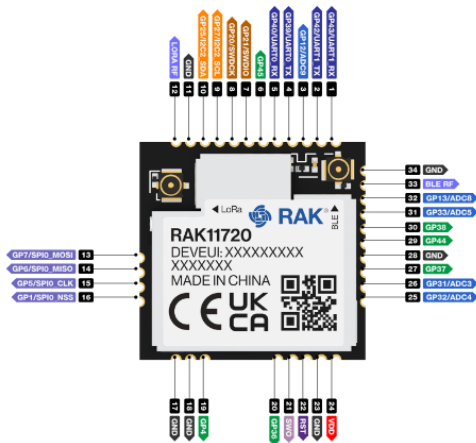


Figure 30. Recommended NRST pin protection





LoRa + BLE WisDuo RAK11720

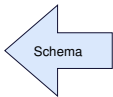


- Operating Voltage VCC: min.1.8 typ.3.3 max.3.6 Volts (V)
- Operating Current:
 - BLE TX Mode **12.7mA** @ 4.0 dBm
 - LORA TX Mode **87mA** @ 20 dBm, 868 MHz
- Sleep current: 2.37µA
- I²C, SPI & UART
- default = **UART0** (AT commands en FW updates)
- default 115200 baud, 8 data bits, 1 stop bit, no parity
- debug pinnen op UART0 (RAK kan ook met pc communiceren ([WisToolBox])(<https://docs.rakwireless.com/product-categories/software-tools/wistoolbox/overview>))
- evt. UART met USB rechtstreeks? (multiplex via jumper pins)
- 2x MHF-4
- [AT commands: RAKwireless Unified Interface V3 (RUI3)](<https://docs.rakwireless.com/product-categories/software-apis-and-libraries/rui3/overview/>)
 - interfaced with an external host via [AT command mode (default)](<https://docs.rakwireless.com/product-categories/software-apis-and-libraries/rui3/at-command-manual/>)
 - custom serial interface can be created by the user using [Custom Mode](<https://docs.rakwireless.com/product-categories/software-apis-and-libraries/rui3/custom-mode/>)
- AT commands ook via I2C en SPI? Of worden deze alleen gebruikt bij custom firmware (RAK als master)?

Antennes: MHF4

Make sure that an antenna is always connected. Using these transceivers

without an antenna can damage the module



GPS-GNSS Quectel LC76GPAMD

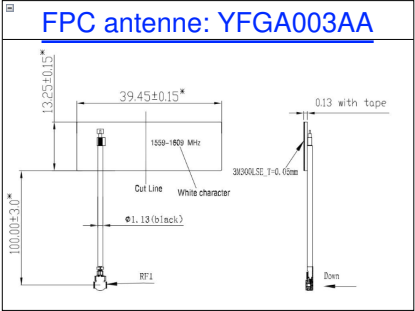
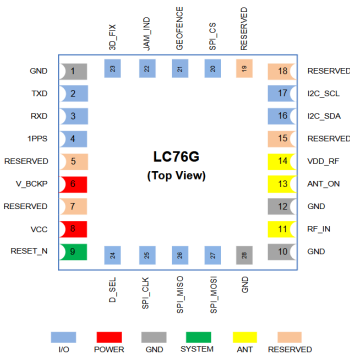
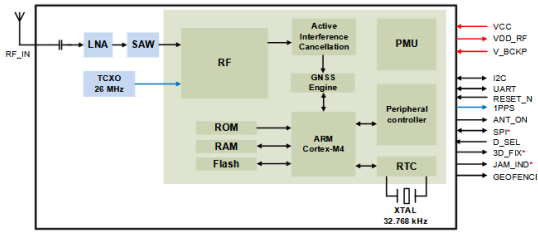
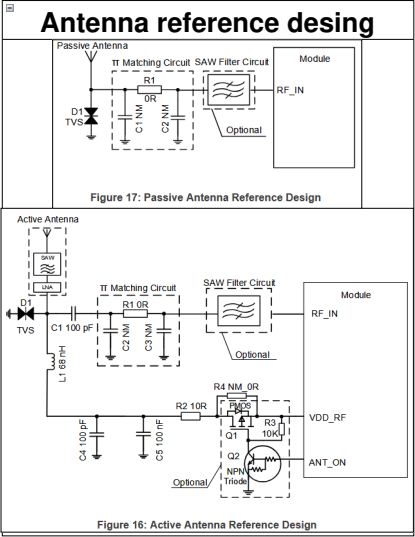
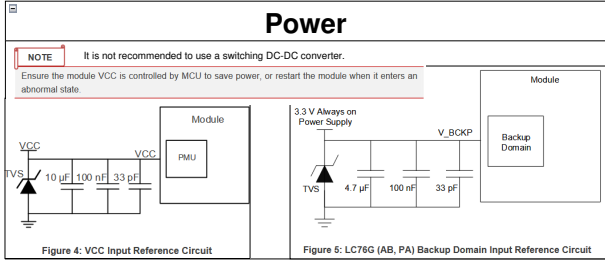
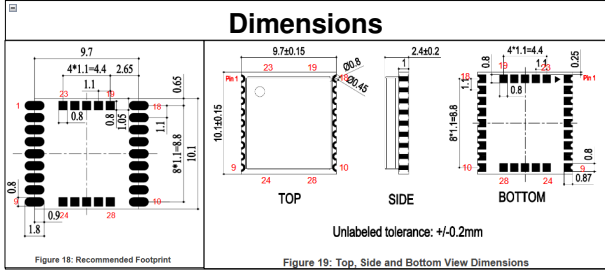
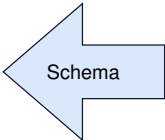


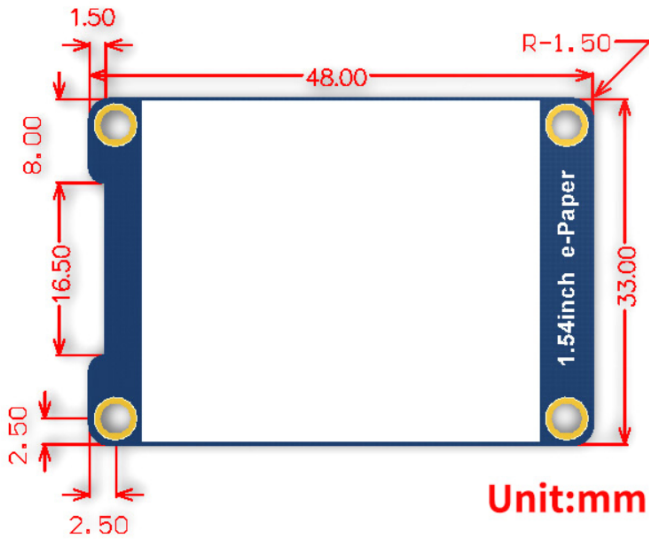
Table 8: Interface Selection by D_SEL			
Voltage Level of D_SEL upon Module Startup	Interface	I2C	SPI
Low	UART	Can be used for communication and Can be used for firmware upgrade (in communication Download mode)	
High		Can be used for communication	Can be used for communication and firmware upgrade (in Download mode)

- Power (default constellations)
Normal Operation:
*10 mA** (33 mW) @ Acquisition
10 mA (33 mW) @ Tracking
Power Saving Modes:
5.5 mA (18.15 mW) @ ALP Mode
13 μ A (42.9 μ W) @ Backup Mode
- Power (GPS + GLONASS):
Normal Operation:
9 mA (29.7 mW) @ Acquisition
9 mA (29.7 mW) @ Tracking
Power Saving Modes:
4.6 mA (15.18 mW) @ ALP Mode

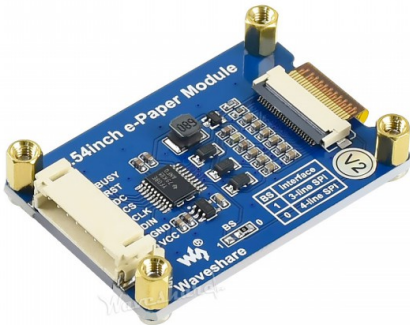


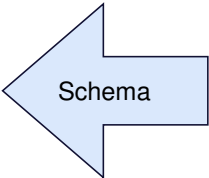


ePaper
Waveshare 1.54" module

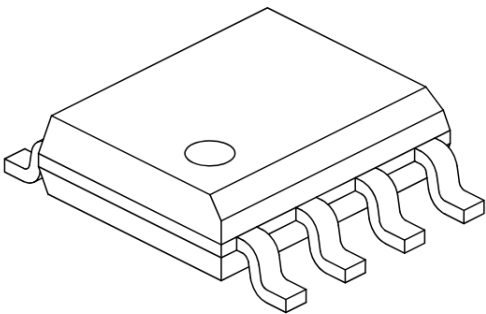
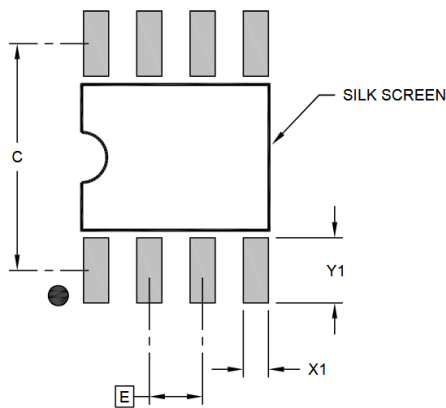


ePaper module STM32 (demo)		
VCC	3.3V	3.3V/5V
GND	GND	
DIN	PA7	SPI MOSI pin
CLK	PA5	SPI SCK pin
CS	PA4	SPI chip selection, low active
DC	PA2	Data/Command selection (high for data, low for command)
RST	PA1	External reset, low active
BUSY	PA3	Busy status output, high active





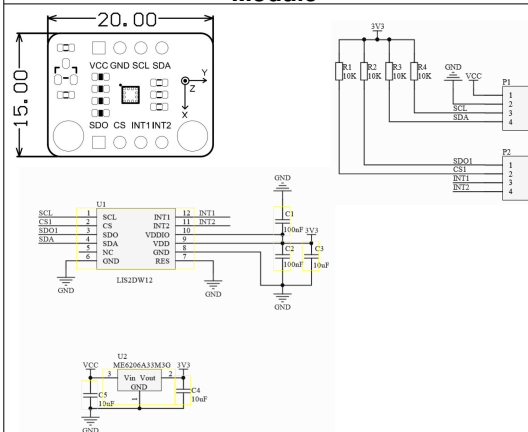
ECC secure element [ATECC608B-SSHDA-T](#)



Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	1.27 BSC		
Contact Pad Spacing	C		5.40	
Contact Pad Width (X8)	X1			0.60
Contact Pad Length (X8)	Y1			1.55

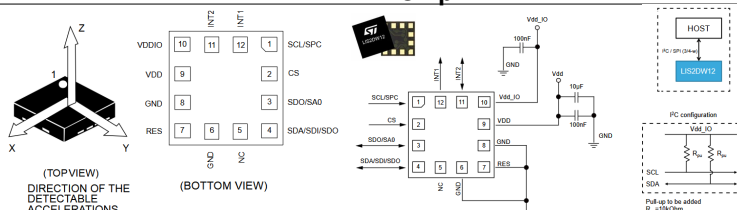
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Pins	N	8		
Pitch	e	1.27 BSC		
Overall Height	A	-	-	1.75
Molded Package Thickness	A2	1.25	-	-
Standoff	§	0.10	-	0.25
Overall Width	E	6.00 BSC		
Molded Package Width	E1	3.90 BSC		
Overall Length	D	4.90 BSC		
Chamfer (Optional)	h	0.25	-	0.50
Foot Length	L	0.40	-	1.27
Footprint	L1	1.04 REF		
Foot Angle	φ	0°	-	8°
Lead Thickness	c	0.17	-	0.25
Lead Width	b	0.31	-	0.51
Mold Draft Angle Top	α	5°	-	15°
Mold Draft Angle Bottom	β	5°	-	15°

Module



No.	Silkscreen	Description
1	VCC	5V/3V3
2	GND	GND
3	SCL	I2C Clock Line/SPI Clock Line
4	SDA	I2C Data Line/SPI Data Line
5	INT2	Interrupt Pin 2
6	INT1	Interrupt Pin 1
7	CS	SPI chip-select line
8	SDO	I2C Address select pin/spi/data line MOSI

Chip

**Table 1. Pin description**

Pin#	Name	Function
1	SCL	PC serial clock (SCL)
	SPC	SPI serial port clock (SPC)
2 ⁽¹⁾	CS	Enable SPI PC/SPI mode selection (1: SPI idle mode / PC communication enabled; 0: SPI communication mode / PC disabled)
3 ⁽¹⁾	SDO	SPI serial data output (SDO)
	SA0	PC less significant bit of the device address (SA0)
4	SDA	PC serial data (SDA)
	SDI	SPI serial data input (SDI)
	SDO	3-wire interface serial data output (SDO)
5	NC	Internally not connected. Can be tied to VDD, VDDIO, or GND.
6	GND	0 V supply
7	RES	Connect to GND
8	GND	0 V supply
9	VDD	Power supply
10	VDD_IO	Power supply for I/O pins
11	INT2	Interrupt pin 2. Clock input when selected in single data conversion on demand.
12	INT1	Interrupt pin 1.

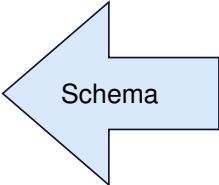
Table 2. Internal pull-up values (typ.) for SDO/SA0 and CS pins

Vdd_IO	Resistor value for SDO/SA0 and CS pins
	Typ. (k Ω)
1.7 V	54.4
1.8 V	49.2
2.5 V	30.4
3.6 V	20.4

The registers embedded inside the LIS2DW12 may be accessed through both the I²C and SPI serial interfaces. The latter may be software configured to operate either in 3-wire or 4-wire interface mode. The serial interfaces are mapped to the same pins. To select/exploit the I²C interface, the CS line must be tied high (that is, connected to V_{DD} IO).

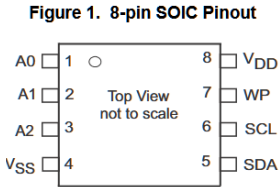
Table 14. Serial interface pin description

Pin name	Pin description
	Enable SPI
CS	PC/SPI mode selection (1: SPI clock mode / PC communication enabled, 0: SPI communication mode / PC disabled)
SCL	SPI serial clock (SCL)
SPC	SPI serial port clock (SPC)
SDA	PC serial data (SDA)
SDI	SPI serial data input (SDI)
SDO	3-wire interface serial data output (SDO)
SA0	PC address selection (SA0)
S00	SPI serial data output (SDO)

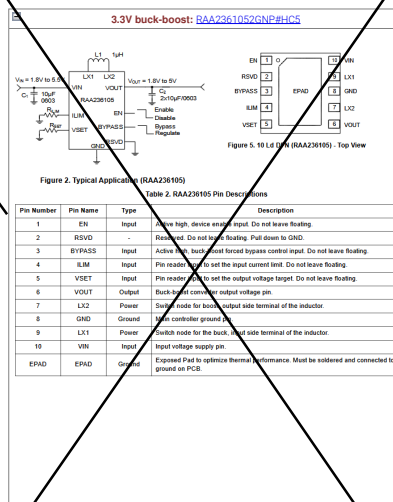
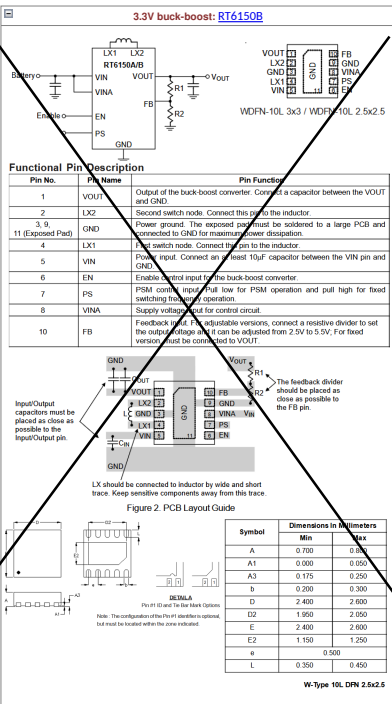


FRAM

FM24V01A-GTR



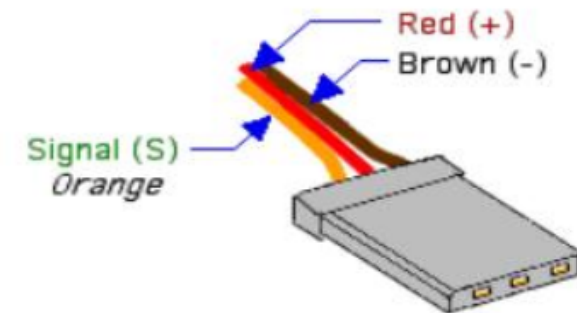
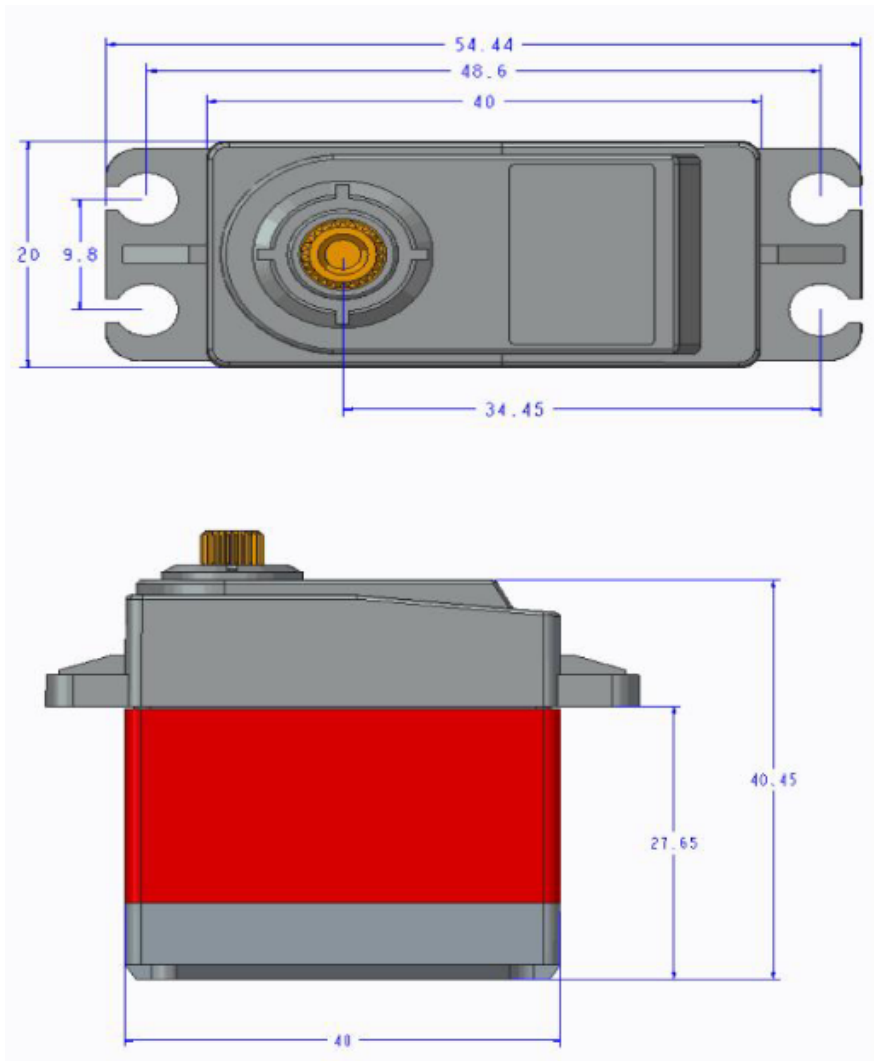
Pin Name	I/O Type	Description
A0–A2	Input	Device Select Address 0–2. These pins are used to select one of up to eight devices of the same type on the same two-wire bus. To select the device, the address value on the three pins must match the corresponding bits contained in the slave address. The address pins are pulled down internally.
SDA	Input/Output	Serial Data Address. This is a bidirectional pin for the two-wire interface. It is open-drain and is intended to be wire-AND'd with other devices on the two-wire bus. The input buffer incorporates a Schmitt trigger for noise immunity and the output driver includes slope control for falling edges. An external pull-up resistor is required.
SCL	Input	Serial Clock. The serial clock pin for the two-wire interface. Data is clocked out of the part on the falling edge, and into the device on the rising edge. The SCL input also incorporates a Schmitt trigger input for noise immunity.
WP	Input	Write Protect. When tied to V _{DD} , addresses in the entire memory map will be write-protected. When WP is connected to ground, all addresses are write enabled. This pin is pulled down internally.
V _{SS}	Power supply	Ground for the device. Must be connected to the ground of the system.
V _{DD}	Power supply	Power supply input to the device.



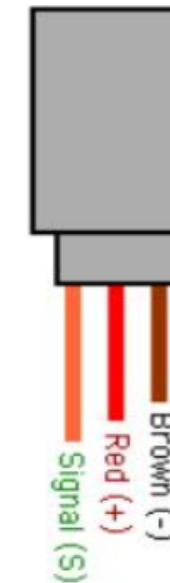
Schema

Servo motor TD-8120MG

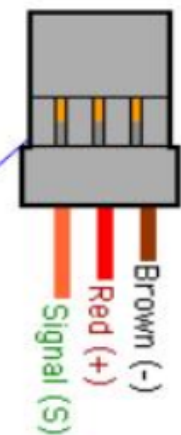
Command signal: Pulse width modification
High = 3 - 5V -> no level shifter needed



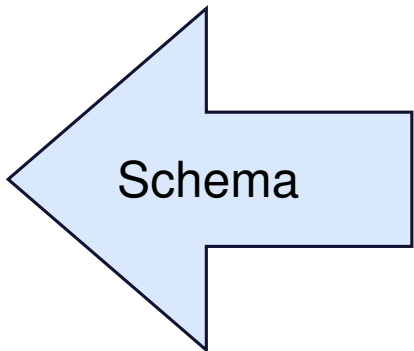
"F" Connector



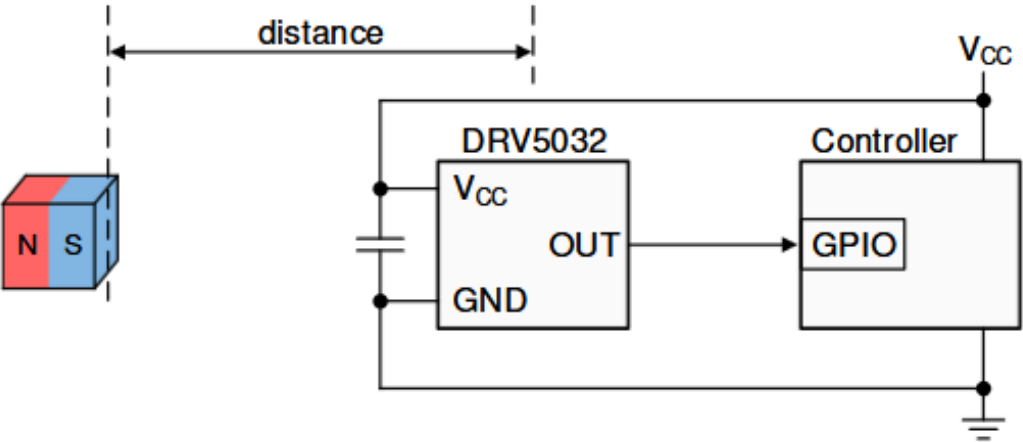
"M" Connector



Lift tabs with knife edge
to release wire pin



Hall sensors (2) [DRV5032DULPGM](#)



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Figure 8-1. Typical Application Diagram

8.5.2 Layout Examples

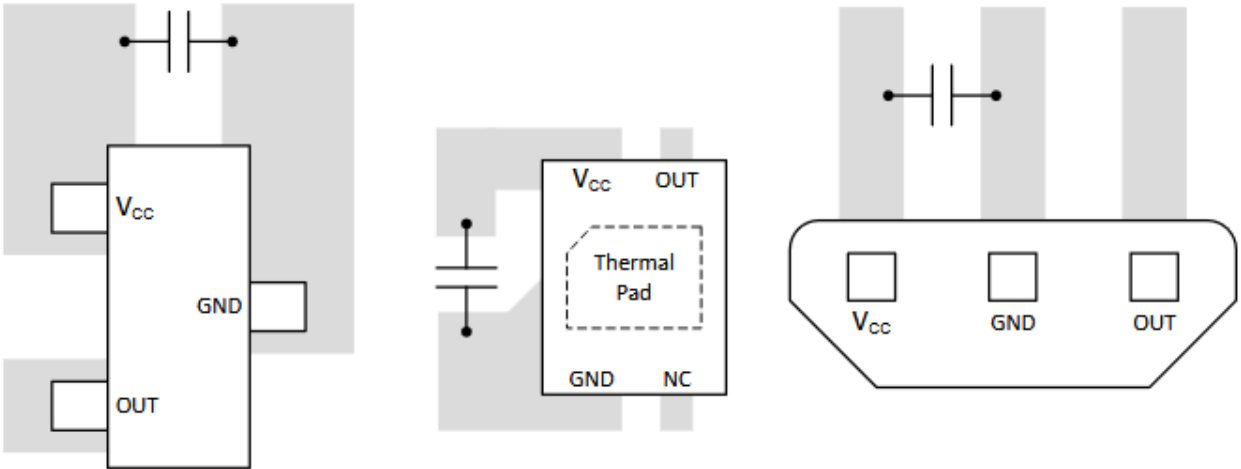


Figure 8-7. Layout Examples

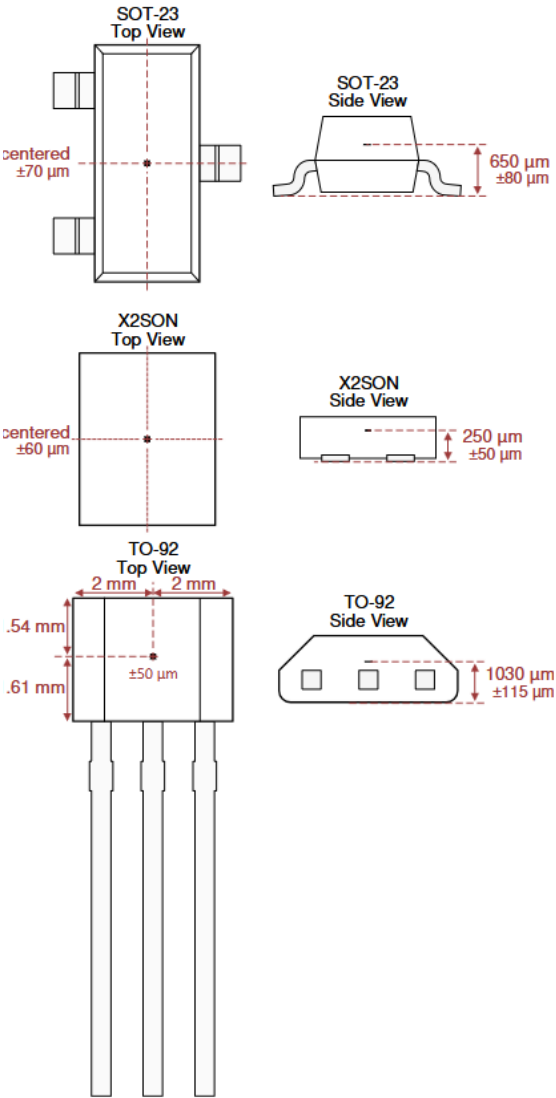
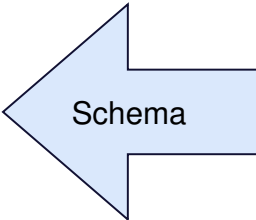
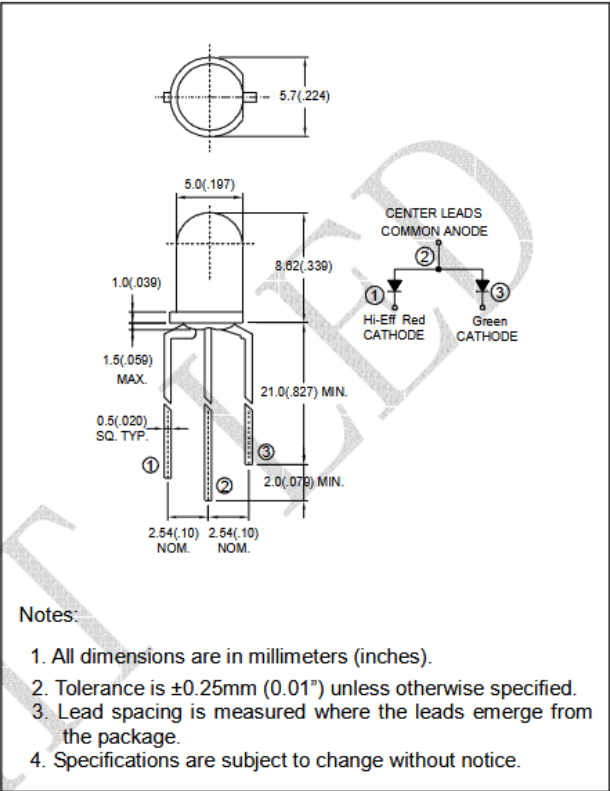


Figure 7-8. Hall Element Location



LED: [BL-BEG204-7-E](#)

● **Package dimensions:**



Parameter	Symbol	Condition	Color	Min.	Typ.	Max.	Unit
Forward Voltage	V_F	$I_F=20\text{mA}$	Hi-Eff Red Green	-	2.0 2.2	2.6 2.6	V

STLINK-V3MINIE

STDC14 connector (CN4)

The STDC14 connector (CN4) enables the connection to an STM32 target using the JTAG or SWD protocol, respecting from pin 3 to pin 12 the MIPI10 (Arm® Cortex® debugging connector) pinout. It also advantageously provides two UART signals for the Virtual COM port. The related pinout for the STDC14 connector is listed in Table 4.

Figure 6. STDC14 connector (CN4) top view

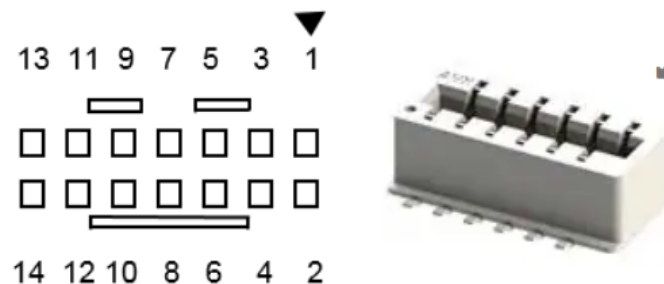


Table 4. STDC14 connector (CN4) pinout

STDC14 pin number	MIPI10 pin number	Pin description	Type
1	-	Reserved ⁽¹⁾	-
2	-	Reserved ⁽¹⁾	-
3	1	T_VCC	I
4	2	T_JTMS/T_SWDIO	I/O
5	3	GND	S
6	4	T_JCLK/T_SWCLK	O
7	5	GND	S
8	6	T_JTDO/T_SWO ⁽²⁾	I
9	7	Reserved ⁽¹⁾	-
10	8	T_JTDI/NC ⁽³⁾	O
11	9	GNDDetect	O
12	10	T_Nrst	O
13	-	T_VCP_RX	O
14	-	T_VCP_TX	I

1. Do not connect on target.

2. SWO is optional, required only for Serial Wire Viewer (SWV) trace.

3. NC means is not required for SWD connection.

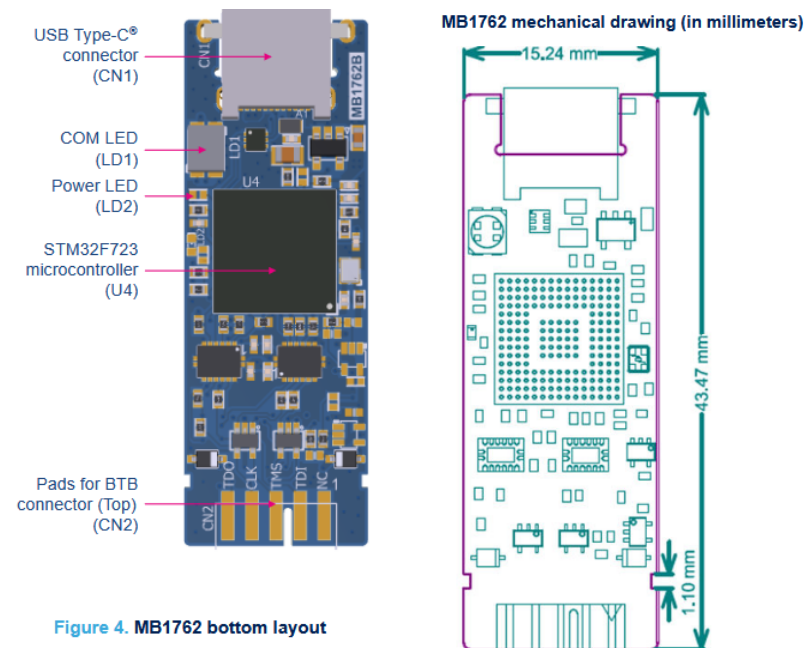
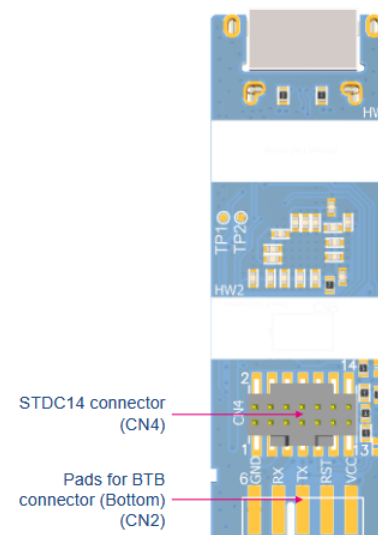


Figure 4. MB1762 bottom layout



Schema

SD adapter module

An SD card adapter that works with SPI. The module works with with a 3.5V of 5V supply and needs to be controlled with 3.3V signals. The pins are shown on the PCB:

Pin	Description
(2x) GND	Ground, min
+3.3	Direct +3.3V supply*
+5	+5V supply that is converted to +3.3V*
CS	3.3V CS signal
MOSI	3.3V MOSI signal
SCK	3.3V SCK signal
MISO	3.3V MISO signal

* You may never use the +3.3V and +5V connected at the same time, just one of the two.



USB4105-GF-A-060

